

Coordinate descent Lasso

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1 Cost function

With Lasso, the cost function is

$$RSS(w) + \lambda \|w\|_1$$

2 Coordinate descent Lasso for normalized features

Pls note $\hat{w}_{-j}$ = the weights without w_j
 $\underline{h}_j$ = the normalized jth feature

We need to compute ρ_j :

$$\rho_j = \sum_{i=0}^{N-1} \underline{h}_j(x_i)(y_i - \hat{y}_i(\hat{w}_{-j}))$$

We set $\hat{w}_j$ as

$$\hat{w}_j = \begin{cases} \rho_j + \frac{\lambda}{2} & \text{if } \rho_j < -\frac{\lambda}{2}, \\ 0 & \text{if } -\frac{\lambda}{2} \leq \rho_j \leq \frac{\lambda}{2}, \\ \rho_j - \frac{\lambda}{2} & \text{if } \rho_j > \frac{\lambda}{2}. \end{cases}$$

But: our convention is that the 0th coefficient (intercept) is not subject to Lasso penalty. To account for this convention, the update to $\hat{w}_0$ is simply:

$$\hat{w}_0 = \rho_0$$

3 Coordinate descent Lasso for unnormalized features

For each feature h_j calculate:

$$Z_j = \sqrt{\sum_{i=0}^{N-1} h_j(x_i)^2}$$

We need to compute ρ_j :

$$\rho_j = \sum_{i=0}^{N-1} h_j(x_i)(y_i - \hat{y}_i(\hat{w}_{-j}))$$

We set $\hat{w}_j$ as

$$\hat{w}_j = \begin{cases} (\rho_j + \frac{\lambda}{2})/Z_j & \text{if } \rho_j < -\frac{\lambda}{2}, \\ 0 & \text{if } -\frac{\lambda}{2} \leq \rho_j \leq \frac{\lambda}{2}, \\ (\rho_j - \frac{\lambda}{2})/Z_j & \text{if } \rho_j > \frac{\lambda}{2}. \end{cases}$$

But for 0th coefficient (intercept):

$$\hat{w}_0 = \rho_0$$

Please note that computation of ρ_j is different between these two situations. In the second situation it uses unnormalized features, and it uses $\hat{y}_i$ calculated with the unnormalized features.